Dongseob Kim

Software Engineer at Samsung Research
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INTERESTS

- Building and designing efficient systems for AI and data platforms
- Scalable data engineering and database technologies.
- Efficient machine learning training and inference

EDUCATION

Seoul National University Seoul, Korea

M.S. (Expected), Graduate School of Data Science Mar. 2025 - Feb. 2027

- Research Assistant at VLDB Lab (Advisor: Professor Sang-won Lee)

Korea University Seoul, Korea

Bachelor of Computer Science and Engineering Mar. 2014 - Feb. 2020

- Total GPA: 3.79/4.5 (Major GPA: 4.06/4.5)

WORK EXPERIENCE

Samsung Research (Samsung Electronics) Seoul, Korea

Software Engineer, Data Intelligence Team Feb. 2020 – Present

- Developed **Personal Data Engine** for Galaxy AI, an on-device data platform enabling personalized data processing and retrieval on mobile devices.
    - Implemented end-to-end data pipeline and search functionality with query optimization.
    - Product was showcased at **Galaxy Unpacked 2025,** recognized with the **Samsung Software Award** (team award).
- Collaborated with research institutes and global startups on applied R&D and product development
    - Contributed to a PoC project with an Israel-based startup.
    - Engaged with Samsung-acquired startup **Oxford Semantic Technologies** to translate requirements into technical specifications.
    - Coordinated a joint project with Samsung R&D Poland and Samsung AI Center Cambridge.
- Researched a distributed deep learning training system for heterogeneous GPU clusters
    - Achieved up to 8.4x speedup and published research paper

PUBLICATIONS

Published Papers

MatSQL: Accelerating Text-to-SQL via Database Schema Materialization

Kyong-Shik Lee*, **Dongseob Kim***, Sang-Won Lee

*ICDE 2026 Workshop (DBML) | *Equal contribution*

언어 모델의 키-값 캐시 재사용을 통한 이미지 질의응답 성능 개선에 관한 연구

Dongseob Kim, Kunwoo Shin, Sangwon Lee

KCC 2025, pp. 1757–1759 | July 2025

Metis: Fast Automatic Distributed Training on Heterogeneous GPUs

Taegeon Um, Byungsoo Oh, Minyoung Kang, Woo-yeon Lee, Goeun Kim, **Dongseob Kim**, Youngtaek Kim, Mohd Muzzammil, Myoungjae Jeon

USENIX ATC 2024, pp. 563–578 | July 2024

Under Review and In Preparation

Decoupling Granularity in Vector Database for RAG

Working title | Under review at VLDB | First author

Adaptive HNSW for Vector Databases

Working title | Manuscript in preparation | Second author

PATENTS

Electronic device and controlling method of electronic device — US Patent 12,307,209 B2 (Granted)

Dongseob Kim, Dohyoung Mo — May. 2025

- Method for recognizing sentences across separately inputted messages in an AI chatbot

Electronic apparatus and control method thereof — US 2023/0 054 108 A1 (Published)

Dongseob Kim — Feb. 2023

- Method for inferring human relationships by analyzing data on mobile devices

HONORS AND AWARDS

Samsung Software Awards - Project Division (Team)

Samsung Electronics — Oct. 2024

Samsung Software Awards - Programming Division (Individual)

Samsung Electronics — Oct. 2023

Excellence Award at Academic Competition for Undergraduates, KCC 2019

Korean Institute of Information Scientists and Engineers — June. 2019

Encouragement Award at Smart Tourism App Competition (Top 15 out of 182)

Korea Tourism Organization, SK Telecom — Dec. 2018

TECHNICAL SKILLS

Advanced — Kotlin, Python, C++, Android, SQL, PyTorch, Docker

Moderate — Java, Kubernetes, GCP(BigQuery, GKE, Dataflow, Bigtable), Web Development

LANGUAGE TEST SCORES

TEPS: 375 (expired) — May 2024

OPIc (English): Advanced Low (expired) — October 2019